

CHITRANG POTDAR

Backend Engineer · Applied AI

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PROFESSIONAL SUMMARY

Backend Engineer with strong expertise in Python, FastAPI/Flask, and REST API design, backed by solid grounding in PostgreSQL, Redis, and asynchronous task processing with Celery. Skilled in applied AI — LLM integration, RAG pipelines, agentic workflows (LangChain, LangGraph, MCP), and AI-assisted backend tooling. Comfortable across the full stack of backend concerns: multi-tenant architecture, testing, observability (Grafana/Loki), and CI/CD. Adept at translating ambiguous requirements into clean, production-grade systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Flask, REST APIs, Celery

Databases & Caching: PostgreSQL, Redis, MSSQL

AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP, AWS Transcribe, Google Cloud STT, Whisper

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD

Observability: Grafana, Loki, Promtail, LangSmith

Testing & Automation: Pytest, Playwright

ML & Data: Scikit-learn, MLflow, ZenML, XGBoost

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune

Oct 2025 – Present

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Grafana/Loki/Promtail · Pytest · Playwright

- Built query-level data isolation across users, portfolios, and organizations on a shared PostgreSQL schema, with Redis-backed session caching to cut redundant DB round-trips; maintained 80%+ Pytest coverage.
- Developed REST APIs for portfolio review, rejected-order tracking, financial calculators, and a CRM module (contacts, deal pipeline, notes, search) for tracking existing clients and new business connections; integrated the Jira API to auto-create issues from application events.
- Built Playwright-based automation for NSE rejected-order processing and client workflows, saving 2–3 hours/week of manual effort; implemented async email delivery via Redis + Celery with UI-visible failure notifications.
- Stood up Grafana dashboards fed by Promtail/Loki for service health and error visibility, and built a domain-specific MCP server exposing investment platform data to LLM clients via business-term-annotated tool definitions.

AI Meeting Intelligence Pipeline

AWS Lambda · API Gateway · S3 · AWS Transcribe · Claude API

- Designed an async serverless pipeline (S3 → AWS Transcribe → Claude API) to work around API Gateway's 29-second timeout, producing structured meeting summaries, agenda scoring, and action items.
- Decoupled ingestion from processing so transcription and LLM analysis run independently of the request/response cycle, keeping the pipeline reliable under Lambda's execution constraints.

AI Voice Conversational System with RAG

FastAPI · Google Cloud STT/Whisper · TTS · Claude API · LlamaIndex

- Built a voice AI pipeline (STT → RAG retrieval via semantic chunking and embedding lookup → LLM response → TTS), grounding answers in a domain knowledge base.
- Designed the retrieval layer as a separable module so the knowledge base can be updated independently of the core processing logic.

Python Developer Intern · ArrayPointer Pvt. Ltd., Pune

Dec 2023 – Apr 2024

- Automated structured data extraction from PDFs (electricity bills, CAS files) using PyPDF2, Camelot, and Tabula, cutting manual extraction effort by ~70% with JSON output.
- Built an XGBoost revenue forecasting model with incremental learning support, improving forecast accuracy by ~20% as new data was ingested.

PROJECTS

SOLID-Agent-Harness — Multi-Agent Code Analysis

Jun 2026

[github.com/Dev-Chitrang/SOLID-Agent-Harness](#) · *LangChain · LangGraph · JavaScript*

- Built a registry-driven CLI running 6 LangGraph workflows (SOLID audit, bug sweep, docs, README gen, review, explain) plus 5 composite pipelines that fan workflows out in parallel with a barrier-joined summary node.
- Designed an Agent → Critic → conditional-retry loop (score-gated, max 3 iterations) using LangGraph's StateGraph, replacing earlier thin-wrapper graphs with patterns that justify a graph framework.
- Built a provider abstraction (OpenAI, Anthropic, Gemini, NVIDIA) swappable without touching graph code, plus document-aware generate/update nodes and diff-aware writes that skip unchanged output.
- Diagnosed a Node.js 18 incompatibility (transitive dependency using an ES2024 regex feature) across the CI matrix, traced it to ora/inquirer majors, and resolved it by baselining to Node 20+.
- Maintained CI (GitHub Actions, Node 20/22 matrix) and a Vitest suite covering core infrastructure and the provider layer.

TaskHub — Multi-Tenant Project Management Backend

Aug – Sep 2025

[github.com/Dev-Chitrang/TaskHub-Full-Stack-Project](#) · *FastAPI · PostgreSQL · Docker · GitHub Actions · Pytest*

- Built a SaaS-style backend with workspace isolation, RBAC, and a complete auth flow (email verification, TOTP-based 2FA, password recovery).
- Containerized with Docker and deployed via a GitHub Actions CI/CD pipeline, with Pytest coverage across all core API modules.

Retail DataOps & MLOps Platform

Mar – Jul 2025

[github.com/Dev-Chitrang/data-warehouse-to-mlops](#) · *FastAPI · MSSQL · Scikit-learn · ZenML · MLflow · NeuralProphet*

- Designed medallion-architecture (Bronze/Silver/Gold) ETL pipelines on SQL Server for structured retail data processing.
- Exposed demand forecasting, customer segmentation, and recommendation models as FastAPI endpoints, managing the ML lifecycle with MLflow (experiment tracking) and ZenML (orchestration).

EDUCATION

B.E. Computer Engineering — 8.65 CGPA

Jun 2021 – Jun 2025

Savitribai Phule Pune University, Pune

CERTIFICATIONS & PUBLICATIONS

- IBM AI & Cloud — AICTE Certified | VOIS AI & Data Analysis — AICTE Certified
- “Corporate Credit Rating Analysis using AI & Sentiment Analysis,” IJIRCCCE (Impact Factor 8.65), Nov 2024